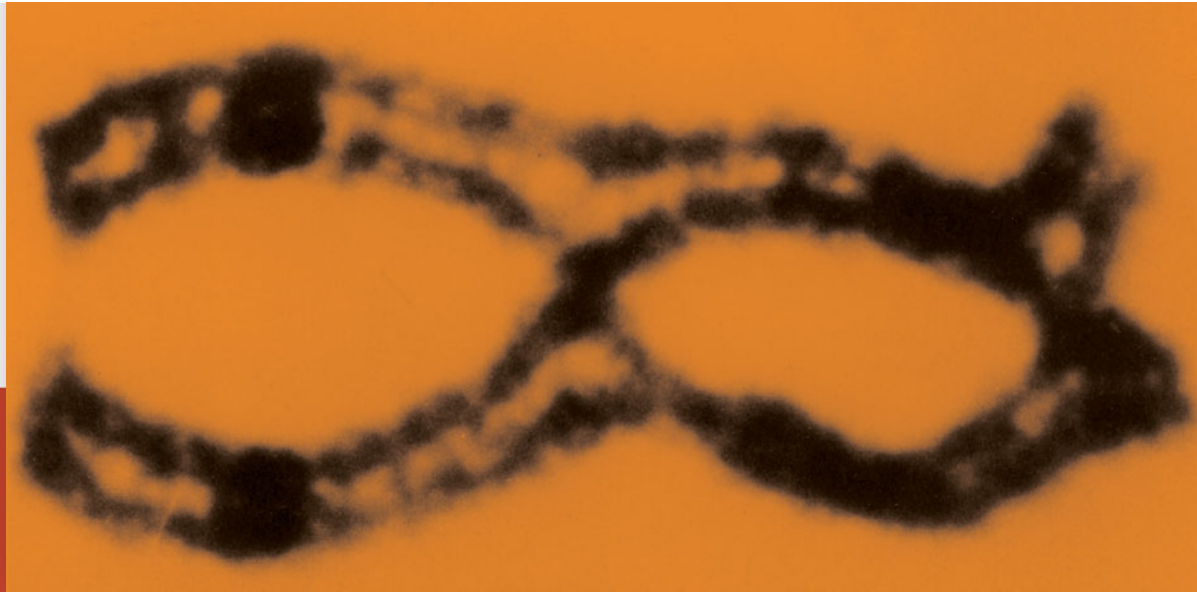


Chiasmata present between synapsed homologs during the first meiotic prophase.



5

Chromosome Mapping in Eukaryotes

CHAPTER CONCEPTS

- Chromosomes in eukaryotes contain large numbers of genes, whose locations are fixed along the length of the chromosomes.
- Unless separated by crossing over, alleles on the same chromosome segregate as a unit during gamete formation.
- Crossing over between homologs during meiosis creates recombinant gametes with different combinations of alleles that enhance genetic variation.
- Crossing over between homologs serves as the basis for the construction of chromosome maps. The greater the distance between two genes on a chromosome, the higher the frequency of crossing over is between them.
- Recombination also occurs between mitotic chromosomes and between sister chromatids.
- Linkage analysis and mapping can be performed for haploid organisms as well as diploid organisms.



alter Sutton, along with Theodor Boveri, was instrumental in uniting the fields of cytology and genetics. As early as 1903, Sutton pointed out the likelihood that there must be many more “unit factors” than chromosomes in most organisms. Soon thereafter, genetic studies with several organisms revealed that certain genes segregate as if they were somehow joined or linked together. Further investigations showed that such genes are part of the same chromosome, and they may indeed be transmitted as a single unit. We now know that most chromosomes contain a very large number of genes. Those that are part of the same chromosome are said to be *linked* and to demonstrate **linkage** in genetic crosses.

Because the chromosome, not the gene, is the unit of transmission during meiosis, linked genes are not free to undergo independent assortment. Instead, the alleles at all loci of one chromosome should, in theory, be transmitted as a unit during gamete formation. However, in many instances this does not occur. As we saw during the first meiotic prophase, when homologs are paired, or synapsed, a reciprocal exchange of chromosome segments may take place (Chapter 2). This **crossing over** results in the reshuffling, or **recombination**, of the alleles between homologs and always occurs during the tetrad stage.

Crossing over is currently viewed as an actual physical breaking and rejoining process that occurs during meiosis. You can see an example in the micrograph that opens this chapter. The exchange of chromosome segments provides an enormous potential for genetic variation in the gametes formed by any individual. This type of variation, in combination with that resulting from independent assortment, ensures that all offspring will contain a diverse mixture of maternal and paternal alleles.

The frequency of crossing over between any two loci on a single chromosome is proportional to the distance between them, known as the **interlocus distance**. Thus, depending on which loci are being considered, the percentage of recombinant gametes varies. This correlation allows us to construct **chromosome maps**, which indicate the relative locations of genes on the chromosomes.

In this chapter, we will discuss linkage, crossing over, and chromosome mapping in more detail. We will also consider a variety of other topics involving the exchange of genetic information, concluding the chapter with the rather intriguing question of why Mendel, who studied seven genes in an organism with seven chromosomes, did not encounter linkage. Or did he?

5.1

Genes Linked on the Same Chromosome Segregate Together

A simplified overview of the major theme of this chapter is given in **Figure 5–1**, which contrasts the meiotic consequences of (a) independent assortment, (b) linkage *without*

crossing over, and (c) linkage *with* crossing over. In **Figure 5–1(a)** we see the results of independent assortment of two pairs of chromosomes, each containing one heterozygous gene pair. No linkage is exhibited. When these same two chromosomes are observed in a large number of meiotic events, they are seen to form four genetically different gametes in equal proportions, each containing a different combination of alleles of the two genes.

Now let's compare these results with what occurs if the same genes are linked on the same chromosome. If no crossing over occurs between the two genes [**Figure 5–1(b)**], only two genetically different kinds of gametes are formed. Each gamete receives the alleles present on one homolog or the other, which is transmitted intact as the result of segregation. This case illustrates **complete linkage**, which produces only **parental**, or **noncrossover**, **gametes**. The two parental gametes are formed in equal proportions. Though complete linkage between two genes seldom occurs, it is useful to consider the theoretical consequences of this concept.

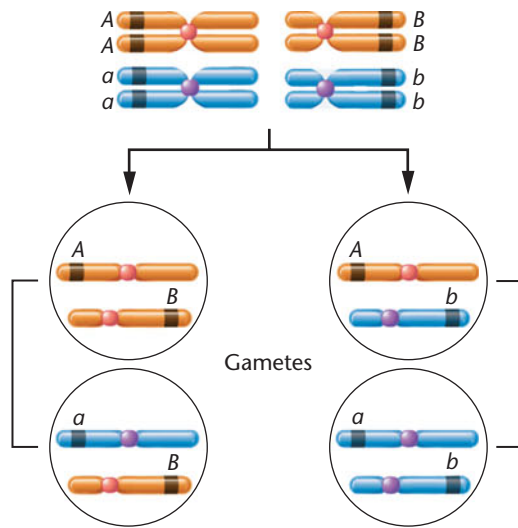
Figure 5–1(c) shows the results when crossing over occurs between two linked genes. As you can see, this crossover involves only two nonsister chromatids of the four chromatids present in the tetrad. This exchange generates two new allele combinations, called **recombinant**, or **crossover**, **gametes**. The two chromatids not involved in the exchange result in noncrossover gametes, like those in **Figure 5–1(b)**. Importantly, the frequency with which crossing over occurs between any two linked genes is proportional to the distance separating the respective loci along the chromosome. As the distance between the two genes increases, the proportion of recombinant gametes increases and that of the parental gametes decreases. In theory, two randomly selected genes can be so close to each other that crossover events are too infrequent to be easily detected. As shown in **Figure 5–1(b)**, this complete linkage produces only parental gametes. On the other hand, if a small, but distinct, distance separates two genes, few recombinant and many parental gametes will be formed.

As we will discuss again later in this chapter, when the loci of two linked genes are far apart, the number of recombinant gametes approaches, but does not exceed, 50 percent. If 50 percent recombinants occur, the result is a 1:1:1:1 ratio of the four types (two parental and two recombinant gametes). In this case, transmission of two linked genes is indistinguishable from that of two unlinked, independently assorting genes. That is, the proportion of the four possible genotypes would be identical, as shown in **Figure 5–1(a)** and **Figure 5–1(c)**.

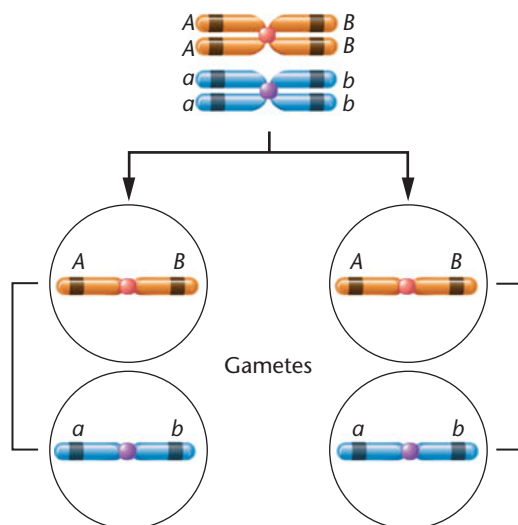
The Linkage Ratio

If complete linkage exists between two genes because of their close proximity, and organisms heterozygous at both loci are mated, a unique F_2 phenotypic ratio results, which

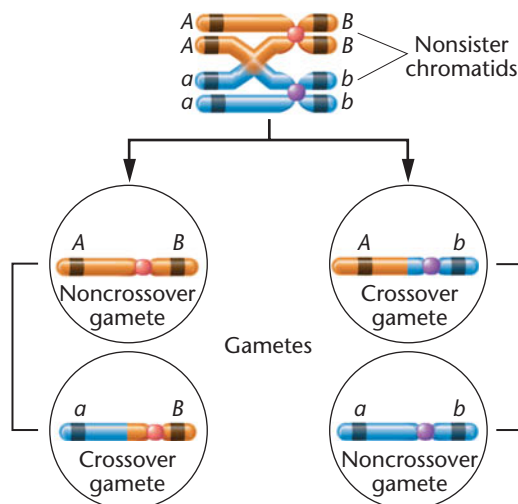
(a) Independent assortment: Two genes on two different homologous pairs of chromosomes



(b) Linkage: Two genes on a single pair of homologs; no exchange occurs



(c) Linkage: Two genes on a single pair of homologs; exchange occurs between two nonsister chromatids



we designate the **linkage ratio**. To illustrate this ratio, let's consider a cross involving the closely linked, recessive, mutant genes *heavy wing vein* (*hv*) and *brown eye* (*bw*) in *Drosophila melanogaster* (Figure 5–2). The normal, wild-type alleles *hv*⁺ and *bw*⁺ are both dominant and result in thin wing veins and red eyes, respectively.

In this cross, flies with normal thin wing veins and mutant brown eyes are mated to flies with mutant heavy wing veins and normal red eyes. In more concise terms, heavy-veined flies are crossed with brown-eyed flies. Linked genes are represented by placing their allele designations (the genetic symbols established in Chapter 4) above and below a single or double horizontal line. Those above the line are located at loci on one homolog, and those below the line are located at the homologous loci on the other homolog. Thus, we represent the P₁ generation as follows:

$$P_1: \frac{hv^+ \ bw}{hv \ bw^+} \times \frac{hv \ bw^+}{hv^+ \ bw}$$

thin, brown heavy, red

Because these genes are located on an autosome, no designation of male or female is necessary.

In the F₁ generation, each fly receives one chromosome of each pair from each parent. All flies are heterozygous for both gene pairs and exhibit the dominant traits of thin veins and red eyes:

$$F_1 = \frac{hv^+ \ bw}{hv \ bw^+}$$

thin, red

As shown in Figure 5–2(a), when the F₁ generation is interbred, each F₁ individual forms only parental gametes because of complete linkage. Following fertilization, the F₂ generation is produced in a 1:2:1 phenotypic and genotypic ratio. One-fourth of this generation shows thin wing veins and brown eyes; one-half shows both wild-type traits, namely, thin veins and red eyes; and one-fourth will show heavy wing veins and red eyes. Therefore, the ratio is 1 heavy: 2 wild: 1 brown. Such a 1:2:1 ratio is characteristic of complete linkage. Complete linkage is usually observed only when genes are very close together and the number of progeny is relatively small.

FIGURE 5–1 Results of gamete formation when two heterozygous genes are (a) on two different pairs of chromosomes; (b) on the same pair of homologs, but with no exchange occurring between them; and (c) on the same pair of homologs, but with an exchange occurring between two nonsister chromatids. Note in this and the following figures that members of homologous pairs of chromosomes are shown in two different colors. This convention was established in Chapter 2 (see, for example, Figure 2–7 and Figure 2–11).

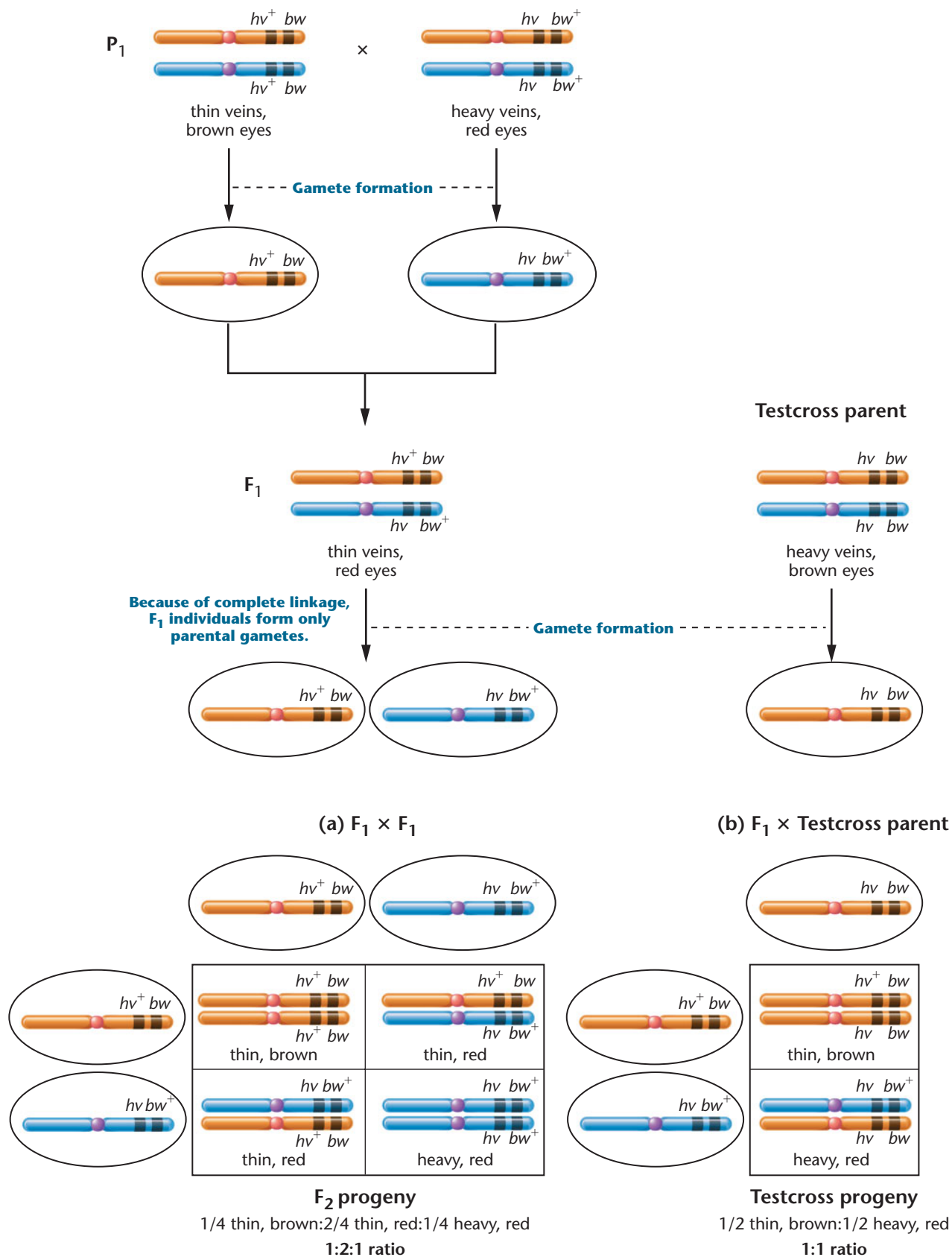


FIGURE 5-2 Results of a cross involving two genes located on the same chromosome and demonstrating complete linkage. (a) The F₂ results of the cross. (b) The results of a testcross involving the F₁ progeny.

Figure 5–2(b) demonstrates the results of a testcross with the F_1 flies. Such a cross produces a 1:1 ratio of thin, brown and heavy, red flies. Had the genes controlling these traits been incompletely linked or located on separate autosomes, the testcross would have produced four phenotypes rather than two.

When large numbers of mutant genes in any given species are investigated, genes located on the same chromosome show evidence of linkage to one another. As a result, **linkage groups** can be identified, one for each chromosome. In theory, the number of linkage groups should correspond to the haploid number of chromosomes. In diploid organisms in which large numbers of mutant genes are available for genetic study, this correlation has been confirmed.

NOW SOLVE THIS

5–1 Consider two hypothetical recessive autosomal genes a and b , where a heterozygote is testcrossed to a double-homozygous mutant. Predict the phenotypic ratios under the following conditions:

- (a) a and b are located on separate autosomes.
- (b) a and b are linked on the same autosome but are so far apart that a crossover always occurs between them.
- (c) a and b are linked on the same autosome but are so close together that a crossover almost never occurs.

■ **HINT:** This problem involves an understanding of linkage, crossing over, and independent assortment. The key to its solution is to be aware that results are indistinguishable when two genes are unlinked compared to the case where they are linked but so far apart that crossing over always intervenes between them during meiosis.

5.2

Crossing Over Serves as the Basis for Determining the Distance between Genes in Chromosome Mapping

It is highly improbable that two randomly selected genes linked on the same chromosome will be so close to one another along the chromosome that they demonstrate complete linkage. Instead, crosses involving two such genes will almost always produce a percentage of offspring resulting from recombinant gametes. The percentage will vary depending on the distance between the two genes along the chromosome. This phenomenon was first explained in 1911 by two *Drosophila* geneticists, Thomas H. Morgan and his undergraduate student, Alfred H. Sturtevant.

Morgan and Crossing Over

As you may recall from our discussion in Chapter 4, Morgan was the first to discover the phenomenon of X-linkage.

In his studies, he investigated numerous *Drosophila* mutations located on the X chromosome. His original analysis, based on crosses involving only one gene on the X chromosome, led to the discovery of X-linked inheritance. However, when he made crosses involving two X-linked genes, his results were initially puzzling. For example, female flies expressing the mutant *yellow* body (y) and *white* eyes (w) alleles were crossed with wild-type males (gray body and red eyes). The F_1 females were wild type, while the F_1 males expressed both mutant traits. In the F_2 the vast majority of the total offspring showed the expected parental phenotypes—yellow-bodied, white-eyed flies and wild-type flies (gray-bodied, red-eyed). The remaining flies, less than 1.0 percent, were either yellow-bodied with red eyes or gray-bodied with white eyes. It was as if the two mutant alleles had somehow separated from each other on the homolog during gamete formation in the F_1 female flies. This outcome is illustrated in cross A of Figure 5–3, using data later compiled by Sturtevant.

When Morgan studied other X-linked genes, the same basic pattern was observed, but the proportion of F_2 phenotypes differed. For example, when he crossed *white-eye*, *miniature-wing* mutants with wild-type flies, only 65.5 percent of all the F_2 flies showed the parental phenotypes, while 34.5 percent of the offspring appeared as if the mutant genes had been separated during gamete formation. This is illustrated in cross B of Figure 5–3, again using data subsequently compiled by Sturtevant.

Morgan was faced with two questions: (1) What was the source of gene separation and (2) why did the frequency of the apparent separation vary depending on the genes being studied? The answer Morgan proposed for the first question was based on his knowledge of earlier cytological observations made by F. A. Janssens and others. Janssens had observed that synapsed homologous chromosomes in meiosis wrapped around each other, creating **chiasmata** (sing. *chiasma*), X-shaped intersections where points of overlap are evident (see the photo on p. 105). Morgan proposed that these chiasmata could represent points of genetic exchange.

Regarding the crosses shown in Figure 5–3, Morgan postulated that if an exchange of chromosome material occurs during gamete formation, at a chiasma between the mutant genes on the two X chromosomes of the F_1 females, the unique phenotypes will occur. He suggested that such exchanges led to recombinant gametes in both the *yellow–white* cross and the *white–miniature* cross, as compared to the parental gametes that underwent no exchange. On the basis of this and other experimentation, Morgan concluded that linked genes are arranged in a linear sequence along the chromosome and that a variable frequency of exchange occurs between any two genes during gamete formation.

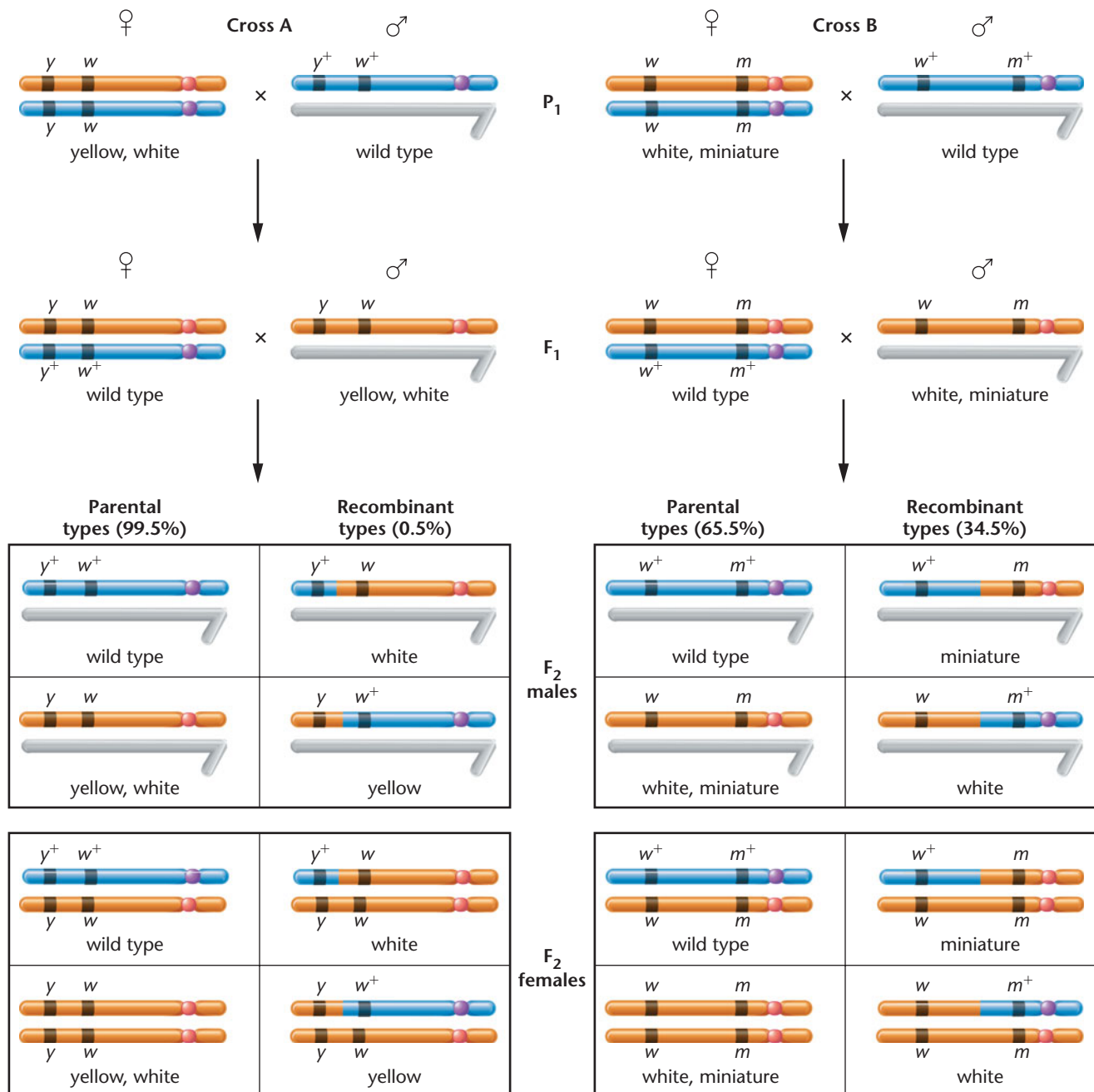


FIGURE 5-3 The F₁ and F₂ results of crosses involving the *yellow* (*y*), *white* (*w*) mutations (cross A), and the *white*, *miniature* (*m*) mutations (cross B), as compiled by Sturtevant. In cross A, 0.5 percent of the F₂ flies (males and females) demonstrate recombinant phenotypes, which express either *white* or *yellow*. In cross B, 34.5 percent of the F₂ flies (males and females) demonstrate recombinant phenotypes, which are either *miniature* or *white* mutants.

In answer to the second question, Morgan proposed that two genes located relatively close to each other along a chromosome are less likely to have a chiasma form between them than if the two genes are farther apart on the chromosome. Therefore, the closer two genes are, the less likely that a genetic exchange will occur between them. Morgan was the first to propose the term *crossing over* to describe the physical exchange leading to recombination.

Sturtevant and Mapping

Morgan's student, Alfred H. Sturtevant, was the first to realize that his mentor's proposal could be used to map the sequence of linked genes. According to Sturtevant,

"In a conversation with Morgan . . . I suddenly realized that the variations in strength of linkage, already attributed by Morgan to differences in the spatial separation

of the genes, offered the possibility of determining sequences in the linear dimension of a chromosome. I went home and spent most of the night (to the neglect of my undergraduate homework) in producing the first chromosomal map.”

Sturtevant, in a paper published in 1913, compiled data from numerous crosses made by Morgan and other geneticists involving recombination between the genes represented by the *yellow*, *white*, and *miniature* mutants. A subset of these data is shown in Figure 5–3. The frequencies of recombination between each pair of these three genes are as follows:

- | | |
|--------------------------------------|-------|
| (1) <i>yellow</i> , <i>white</i> | 0.5% |
| (2) <i>white</i> , <i>miniature</i> | 34.5% |
| (3) <i>yellow</i> , <i>miniature</i> | 35.4% |

Because the sum of (1) and (2) approximately equals (3), Sturtevant suggested that the recombination frequencies between linked genes are additive. On this basis, he predicted that the order of the genes on the X chromosome is *yellow*–*white*–*miniature*. In arriving at this conclusion, he reasoned as follows: The *yellow* and *white* genes are apparently close to each other because the recombination frequency is low. However, both of these genes are quite far from the *miniature* gene because the *white*–*miniature* and *yellow*–*miniature* combinations show larger recombination frequencies. Because *miniature* shows more recombination with *yellow* than with *white* (35.4 percent vs. 34.5 percent), it follows that *white* is located between the other two genes, not outside of them.

Sturtevant knew from Morgan’s work that the frequency of exchange could be used as an estimate of the distance between two genes or loci along the chromosome. He constructed a chromosome map of the three genes on the X chromosome, setting one map unit (mu) equal to 1 percent recombination between two genes.* The distance between *yellow* and *white* is thus 0.5 mu, and the distance between *yellow* and *miniature* is 35.4 mu. It follows that the distance between *white* and *miniature* should be $35.4 - 0.5 = 34.9$ mu. This estimate is close to the actual frequency of recombination between *white* and *miniature* (34.5 mu). The map for these three genes is shown in Figure 5–4. The fact that these numbers do not add up perfectly is due to normal variation that one would expect between crosses, leading to the minor imprecisions encountered in independently conducted mapping experiments.

In addition to these three genes, Sturtevant considered crosses involving two other genes on the X chromosome and

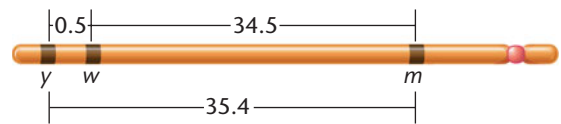


FIGURE 5–4 A map of the *yellow* (*y*), *white* (*w*), and *miniature* (*m*) genes on the X chromosome of *Drosophila melanogaster*. Each number represents the percentage of recombinant offspring produced in one of three crosses, each involving two different genes.

produced a more extensive map that included all five genes. He and a colleague, Calvin Bridges, soon began a search for autosomal linkage in *Drosophila*. By 1923, they had clearly shown that linkage and crossing over are not restricted to X-linked genes but could also be demonstrated with autosomes. During this work, they made another interesting observation. In *Drosophila*, crossing over was shown to occur only in females. The fact that no crossing over occurs in males made genetic mapping much less complex to analyze in *Drosophila*. While crossing over does occur in both sexes in most other organisms, crossing over in males is often observed to occur less frequently than in females. For example, in humans, such recombination occurs only about 60 percent as often in males compared to females.

Although many refinements have been added to chromosome mapping since Sturtevant’s initial work, his basic principles are accepted as correct. These principles are used to produce detailed chromosome maps of organisms for which large numbers of linked mutant genes are known. Sturtevant’s findings are also historically significant to the broader field of genetics. In 1910, the **chromosomal theory of inheritance** was still widely disputed—even Morgan was skeptical of this theory before he conducted his experiments. Research has now firmly established that chromosomes contain genes in a linear order and that these genes are the equivalent of Mendel’s unit factors.

Single Crossovers

Why should the relative distance between two loci influence the amount of crossing over and recombination observed between them? During meiosis, a limited number of crossover events occur in each tetrad. These recombinant events occur randomly along the length of the tetrad. Therefore, the closer that two loci reside along the axis of the chromosome, the less likely that any **single crossover** event will occur between them. The same reasoning suggests that the farther apart two linked loci, the more likely a random crossover event will occur in between them.

In Figure 5–5(a), a single crossover occurs between two nonsister chromatids, but not between the two loci being

*In honor of Morgan’s work, map units are often referred to as centiMorgans (cM).

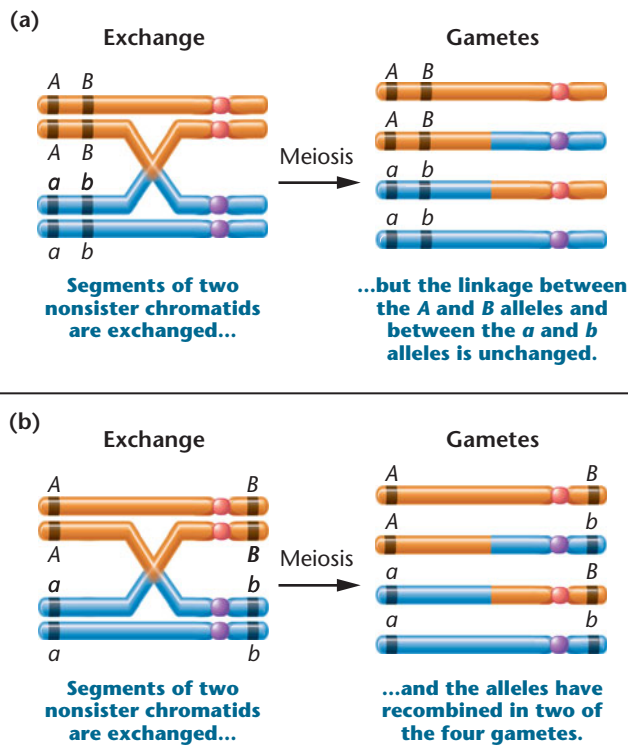


FIGURE 5-5 Two examples of a single crossover between two nonsister chromatids and the gametes subsequently produced. In (a) the exchange does not alter the linkage arrangement between the alleles of the two genes, only parental gametes are formed, and the exchange goes undetected. In (b) the exchange separates the alleles, resulting in recombinant gametes, which are detectable.

studied; therefore, the crossover is undetected because no recombinant gametes are produced for the two traits of interest. In [Figure 5-5\(b\)](#), where the two loci under study are quite far apart, the crossover does occur between them, yielding gametes in which the traits of interest are recombined.

When a single crossover occurs between two nonsister chromatids, the other two chromatids of the tetrad are not involved in the exchange and enter the gamete unchanged. Even if a single crossover occurs 100 percent of the time between two linked genes, recombination is subsequently observed in only 50 percent of the potential gametes formed. This concept is diagrammed in [Figure 5-6](#). Theoretically, if we assume only single exchanges between a given pair of loci and observe 20 percent recombinant gametes, we will conclude that crossing over actually occurs between these two loci in 40 percent of the tetrads. The general rule is that, under these conditions, the percentage of tetrads involved in an exchange between two genes is twice as great as the percentage of recombinant gametes produced. Therefore, the theoretical limit of observed recombination due to crossing over is 50 percent.

When two linked genes are more than 50 map units apart, a crossover can theoretically be expected to occur between them in 100 percent of the tetrads. If this prediction were achieved, each tetrad would yield equal proportions of the four gametes shown in [Figure 5-6](#), just as if the genes were on different chromosomes and assorting independently. For a variety of reasons, this theoretical limit is seldom achieved.

5.3

Determining the Gene Sequence during Mapping Requires the Analysis of Multiple Crossovers

The study of single crossovers between two linked genes provides a basis for determining the *distance* between them. However, when many linked genes are studied, their *sequence* along the chromosome is more difficult to determine. Fortunately, the discovery that multiple crossovers occur between the chromatids of a tetrad has facilitated the

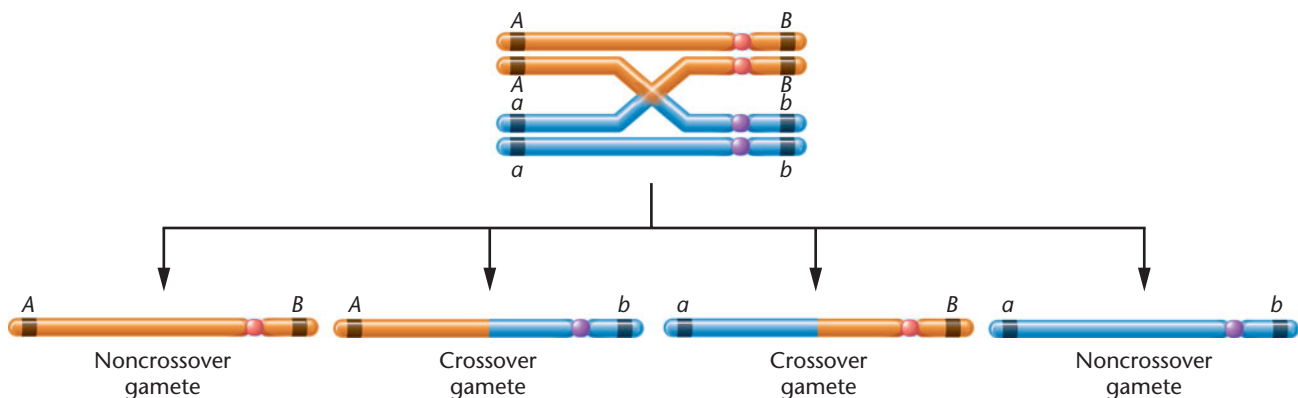


FIGURE 5-6 The consequences of a single exchange between two nonsister chromatids occurring in the tetrad stage. Two noncrossover (parental) and two crossover (recombinant) gametes are produced.

NOW SOLVE THIS

5–2 With two pairs of genes involved (P/p and Z/z), a testcross ($ppzz$) with an organism of unknown genotype indicated that the gametes produced were in the following proportions:

PZ , 42.4%; Pz , 6.9%; pZ , 7.1%; pz , 43.6%

Draw all possible conclusions from these data.

■ **HINT:** *This problem involves an understanding of the proportionality between crossover frequency and distance between genes. The key to its solution is to be aware that noncrossover and crossover gametes occur in reciprocal pairs of approximately equal proportions.*

process of producing more extensive chromosome maps. As we shall see next, when three or more linked genes are investigated simultaneously, it is possible to determine first the sequence of genes and then the distances between them.

Multiple Exchanges

It is possible that in a single tetrad, two, three, or more exchanges will occur between nonsister chromatids as a result of several crossing over events. Double exchanges of genetic material result from **double crossovers (DCOs)**, as shown in Figure 5–7. To study a double exchange, three gene pairs must be investigated, each heterozygous for two alleles. Before we determine the frequency of recombination among all three loci, let's review some simple probability calculations.

As we have seen, the probability of a single exchange occurring in between the A and B or the B and C genes is related directly to the distance between the respective loci.

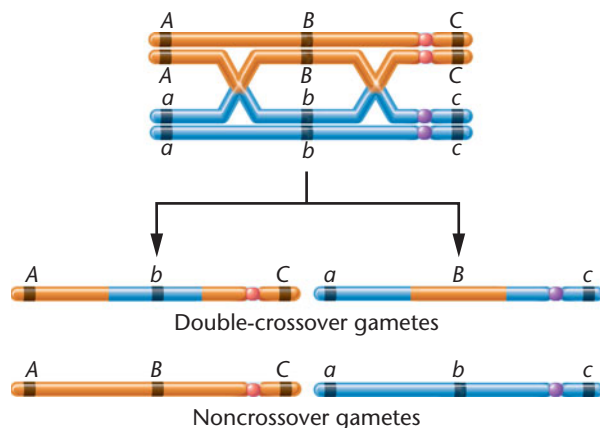


FIGURE 5–7 Consequences of a double exchange occurring between two nonsister chromatids. Because the exchanges involve only two chromatids, two noncrossover gametes and two double-crossover gametes are produced. The Chapter Opening photograph on p. 105 illustrates two chiasmata present in a tetrad isolated during the first meiotic prophase stage.

The closer A is to B and B is to C , the less likely it is that a single exchange will occur in between either of the two sets of loci. In the case of a double crossover, two separate and independent events or exchanges must occur simultaneously. The mathematical probability of two independent events occurring simultaneously is equal to the product of the individual probabilities. This is the *product law* introduced in Chapter 3.

Suppose that crossover gametes resulting from single exchanges are recovered 20 percent of the time ($p = 0.20$) between A and B , and 30 percent of the time ($p = 0.30$) between B and C . The probability of recovering a double-crossover gamete arising from two exchanges (between A and B and between B and C) is predicted to be $(0.20)(0.30) = 0.06$, or 6 percent. It is apparent from this calculation that the expected frequency of double-crossover gametes is always expected to be much lower than that of either single-crossover class of gametes.

If three genes are relatively close together along one chromosome, the expected frequency of double-crossover gametes is extremely low. For example, suppose that the A – B distance in Figure 5–7 is 3 mu and the B – C distance is 2 mu. The expected double-crossover frequency is $(0.03)(0.02) = 0.0006$, or 0.06 percent. This translates to only 6 events in 10,000. Thus in a mapping experiment where closely linked genes are involved, very large numbers of offspring are required to detect double-crossover events. In this example, it is unlikely that a double crossover will be observed even if 1000 offspring are examined. Thus, it is evident that if four or five genes are being mapped, even fewer triple and quadruple crossovers can be expected to occur.

Three-Point Mapping in *Drosophila*

The information presented in the previous section enables us to map three or more linked genes in a single cross. To illustrate the mapping process in its entirety, we examine two situations involving three linked genes in two quite different organisms.

To execute a successful mapping cross, three criteria must be met:

1. The genotype of the organism producing the crossover gametes must be heterozygous at all loci under consideration. If homozygosity occurred at any locus, all gametes produced would contain the same allele, precluding mapping analysis.
2. The cross must be constructed so that the genotypes of all gametes can be accurately determined by observing the phenotypes of the resulting offspring. This is necessary because the gametes and their genotypes can never be observed directly. To overcome this problem, each phenotypic class must reflect the genotype of the gametes of the parents producing it.

3. A sufficient number of offspring must be produced in the mapping experiment to recover a representative sample of all crossover classes.

These criteria are met in the three-point mapping cross of *Drosophila melanogaster* shown in Figure 5–8. In this cross three X-linked recessive mutant genes—yellow body color, white eye color, and echinus eye shape—are considered. To diagram the cross, we must assume some theoretical sequence, even though we do not yet know if it is correct. In Figure 5–8, we initially assume the sequence of the three genes to be *y-w-ec*. If this is incorrect, our analysis shall demonstrate it and reveal the correct sequence.

In the P₁ generation, males hemizygous for all three wild-type alleles are crossed to females that are homozygous for all three recessive mutant alleles. Therefore, the P₁ males are wild type with respect to body color, eye color, and eye shape. They are said to have a *wild-type phenotype*. The females, on the other hand, exhibit the three mutant traits: yellow body color, white eyes, and echinus eye shape.

This cross produces an F₁ generation consisting of females that are heterozygous at all three loci and males that, because of the Y chromosome, are hemizygous for the three mutant alleles. Phenotypically, all F₁ females are wild type, while all F₁ males are yellow, white, and echinus. The genotype of the F₁ females fulfills the first criterion for constructing a map of the three linked genes; that is, it is heterozygous at the three loci and may serve as the source of recombinant gametes generated by crossing over. Note that, because of the genotypes of the P₁ parents, all three of the mutant alleles are on one homolog and all three wild-type alleles are on the other homolog. With other parents, other arrangements would be possible that could produce a heterozygous genotype. For example, a heterozygous female could have the *y* and *ec* mutant alleles on one homolog and the *w* allele on the other. This would occur if one of her parents was yellow, echinus and the other parent was white.

In our cross, the second criterion is met as a result of the gametes formed by the F₁ males. Every gamete contains either an X chromosome bearing the three mutant alleles or a Y chromosome, which does not contain any of the three loci being considered. Whichever type participates in fertilization, the genotype of the gamete produced by the F₁ female will be expressed phenotypically in the F₂ female and male offspring derived from it. As a result, all noncrossover and crossover gametes produced by the F₁ female parent can be determined by observing the F₂ phenotypes.

With these two criteria met, we can construct a chromosome map from the crosses illustrated in Figure 5–8. First, we must determine which F₂ phenotypes correspond to the various noncrossover and crossover categories. To determine the **noncrossover** F₂ phenotypes, we must identify

individuals derived from the parental gametes formed by the F₁ female. Each such gamete contains an X chromosome unaffected by crossing over. As a result of segregation, approximately equal proportions of the two types of gametes, and subsequently their F₂ phenotypes, are produced. Because they derive from a heterozygote, the genotypes of the two parental gametes and the F₂ phenotypes complement one another. For example, if one is wild type, the other is mutant for all three genes. This is the case in the cross being considered. In other situations, if one chromosome shows one mutant allele, the second chromosome shows the other two mutant alleles, and so on. These are therefore called **reciprocal classes** of gametes and phenotypes.

The two noncrossover phenotypes are most easily recognized because they occur in the greatest proportion of offspring. Figure 5–8 shows that gametes (1) and (2) are present in the greatest numbers. Therefore, flies that are yellow, white, and echinus and those that are normal, or wild type, for all three characters constitute the noncrossover category and represent 94.44 percent of the F₂ offspring.

The second category that can be easily detected is represented by the double-crossover phenotypes. Because of their low probability of occurrence, they must be present in the least numbers. Remember that this group represents two independent but simultaneous single-crossover events. Two reciprocal phenotypes can be identified: gamete 7, which shows the mutant traits yellow and echinus, but normal eye color; and gamete 8, which shows the mutant trait white, but normal body color and eye shape. Together these double-crossover phenotypes constitute only 0.06 percent of the F₂ offspring.

The remaining four phenotypic classes fall into two categories resulting from single crossovers. Gametes 3 and 4, reciprocal phenotypes produced by single-crossover events occurring between the yellow and white loci, are equal to 1.50 percent of the F₂ offspring. Gametes 5 and 6, constituting 4.00 percent of the F₂ offspring, represent the reciprocal phenotypes resulting from single-crossover events occurring between the white and echinus loci.

We can now calculate the map distances between the three loci. The distance between *y* and *w*, or between *w* and *ec*, is equal to the percentage of all detectable exchanges occurring between them. For any two genes under consideration, this includes all related single crossovers as well as all double crossovers. The latter are included because they represent two simultaneous single crossovers. For the *y* and *w* genes, this includes gametes 3, 4, 7, and 8, totaling 1.50% + 0.06%, or 1.56 mu. Similarly, the distance between *w* and *ec* is equal to the percentage of offspring resulting from an exchange between these two loci: gametes 5, 6, 7, and 8, totaling 4.00% + 0.06%, or 4.06 mu. The map of these three loci on the X chromosome is shown at the bottom of Figure 5–8.

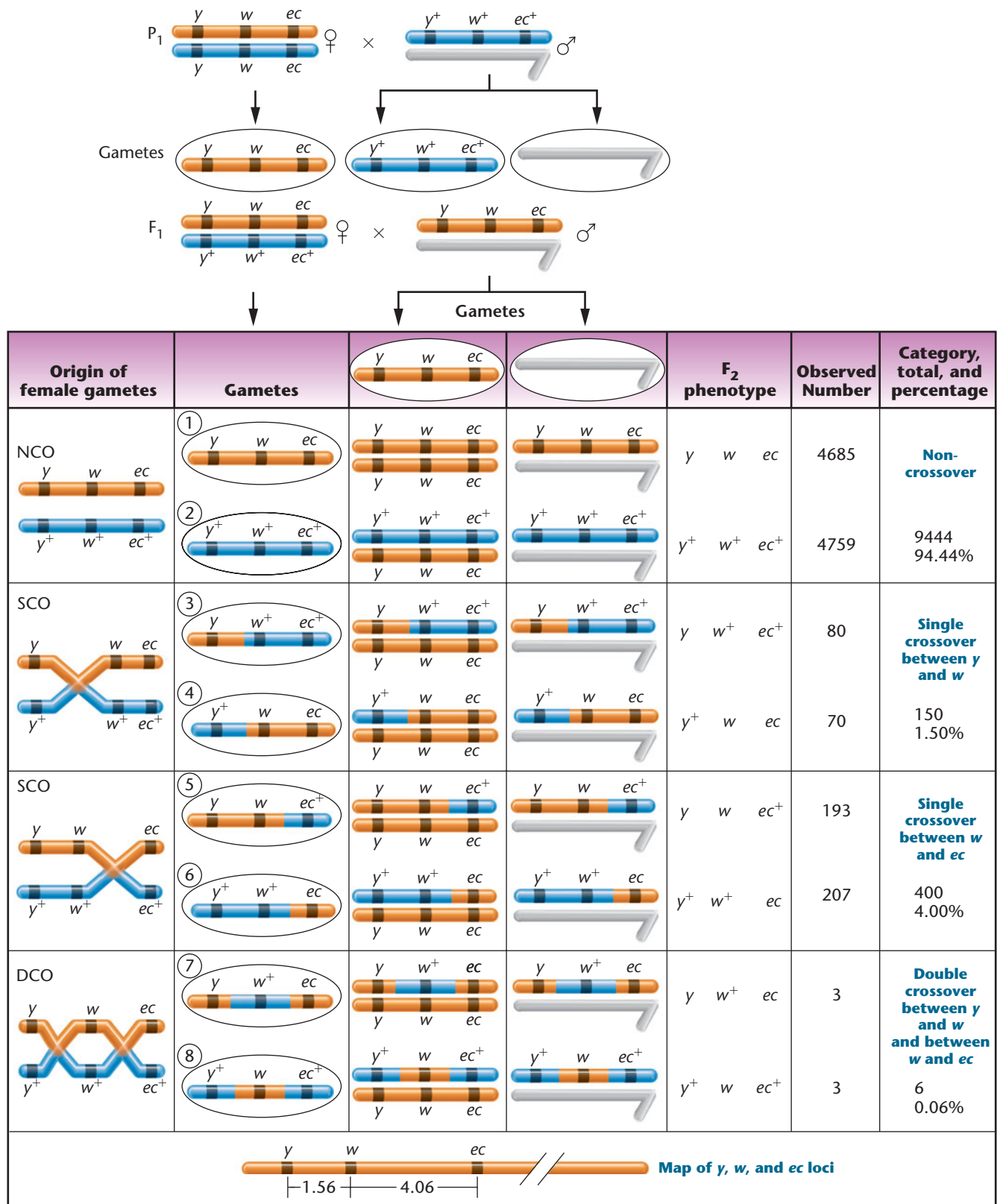


FIGURE 5-8 A three-point mapping cross involving the *yellow* (*y* or *y*⁺), *white* (*w* or *w*⁺), and *echinus* (*ec* or *ec*⁺) genes in *Drosophila melanogaster*. NCO, SCO, and DCO refer to noncrossover, single-crossover, and double-crossover groups, respectively. Centromeres are not drawn on the chromosomes, and only two nonsister chromatids are initially shown in the left-hand column.

Determining the Gene Sequence

In the preceding example, we assumed that the sequence (or order) of the three genes along the chromosome was *y-w-ec*. Our analysis established that the sequence is consistent with the data. However, in most mapping experiments, the gene sequence is not known, and this constitutes another variable in the analysis. In our example, had the gene order been unknown, we could have used one of two methods (which we will study next) to determine it. In your own work, you should select one of these methods and use it consistently.

Method I This method is based on the fact that there are only three possible arrangements, each containing a different one of the three genes between the other two:

- (I) *w-y-ec* (*y* is in the middle)
- (II) *y-ec-w* (*ec* is in the middle)
- (III) *y-w-ec* (*w* is in the middle)

Use the following steps during your analysis to determine the gene order:

1. Assuming any of the three orders, first determine the *arrangement of alleles* along each homolog of the heterozygous parent giving rise to noncrossover and crossover gametes (the *F*₁ female in our example).
2. Determine whether a double-crossover event occurring within that arrangement will produce the *observed double-crossover phenotypes*. Remember that these phenotypes occur least frequently and are easily identified.

3. If this order does not produce the correct phenotypes, try each of the other two orders. One must work!

These steps are shown in **Figure 5–9**, using our *y-w-ec* cross. The three possible arrangements are labeled I, II, and III, as shown above.

1. Assuming that *y* is between *w* and *ec* (arrangement I), the distribution of alleles between the homologs of the *F*₁ heterozygote is:

$$\begin{array}{ccc} w & y & ec \\ \hline w^+ & y^+ & ec^+ \end{array}$$

We know this because of the way in which the *P*₁ generation was crossed: The *P*₁ female contributes an X chromosome bearing the *w*, *y*, and *ec* alleles, while the *P*₁ male contributed an X chromosome bearing the *w*⁺, *y*⁺, and *ec*⁺ alleles.

2. A double crossover within that arrangement yields the following gametes:

$$\underline{w \quad y^+ \quad ec} \quad \text{and} \quad \underline{w^+ \quad y \quad ec^+}$$

Following fertilization, if *y* is in the middle, the *F*₂ double-crossover phenotypes will correspond to these gametic genotypes, yielding offspring that express the white, echinus phenotype and offspring that express the yellow phenotype. Instead, determination of the actual double crossovers reveals them to be yellow, echinus flies and white flies. *Therefore, our assumed order is incorrect.*

Three theoretical sequences	Double-crossover gametes	Phenotypes
I wyec w+y+ec+	w y+ ec w+ y ec+	white, echinus yellow
II yecw y+ec+w+	y ec+ w y+ ec w+	yellow, white echinus
III ywec y+w+ec+	y w+ ec y+ w ec+	yellow, echinus white

FIGURE 5–9 The three possible sequences of the *white*, *yellow*, and *echinus* genes, the results of a double crossover in each case, and the resulting phenotypes produced in a testcross. For simplicity, the two noncrossover chromatids of each tetrad are omitted.

3. If we consider arrangement II, with the ec/ec^+ alleles in the middle, or arrangement III, with the w/w^+ alleles in the middle:

$$(II) \frac{y \quad ec \quad w}{y^+ \quad ec^+ \quad w^+} \quad \text{or} \quad (III) \frac{y \quad w \quad ec}{y^+ \quad w^+ \quad ec^+}$$

we see that arrangement II again provides *predicted* double-crossover phenotypes that do not correspond to the actual (observed) double-crossover phenotypes. The predicted phenotypes are yellow, white flies and echinus flies in the F_2 generation. *Therefore, this order is also incorrect.* However, arrangement III produces the observed phenotypes—yellow, echinus flies and white flies. *Therefore, this arrangement, with the w gene in the middle, is correct.*

To summarize Method I: First, determine the arrangement of alleles on the homologs of the heterozygote yielding the crossover gametes by identifying the reciprocal non-crossover phenotypes. Then, test each of the three possible orders to determine which one yields the observed double-crossover phenotypes—the one that does so represents the correct order. This method is summarized in Figure 5–9.

Method II Method II also begins by determining the arrangement of alleles along each homolog of the heterozygous parent. In addition, it requires one further assumption:

Following a double-crossover event, the allele in the middle position will fall between the outside, or flanking, alleles that were present on the opposite parental homolog.

To illustrate, assume order I, w – y – ec , in the following arrangement:

$$\frac{w \quad y \quad ec}{w^+ \quad y^+ \quad ec^+}$$

Following a double-crossover event, the y and y^+ alleles would be switched to this arrangement:

$$\frac{w \quad y^+ \quad ec}{w^+ \quad y \quad ec^+}$$

After segregation, two gametes would be formed:

$$\underline{w \quad y^+ \quad ec} \quad \text{and} \quad \underline{w^+ \quad y \quad ec^+}$$

Because the genotype of the gamete will be expressed directly in the phenotype following fertilization, the double-crossover phenotypes will be:

white, echinus flies and yellow flies

Note that the *yellow* allele, assumed to be in the middle, is now associated with the two outside markers of the other homolog, w^+ and ec^+ . However, these predicted phenotypes

do not coincide with the observed double-crossover phenotypes. Therefore, the *yellow* gene is not in the middle.

This same reasoning can be applied to the assumption that the *echinus* gene or the *white* gene is in the middle. In the former case, we will reach a negative conclusion. If we assume that the *white* gene is in the middle, the *predicted* and *actual* double crossovers coincide. Therefore, we conclude that the *white* gene is located between the *yellow* and *echinus* genes.

To summarize Method II, determine the arrangement of alleles on the homologs of the heterozygote yielding crossover gametes. Then examine the actual double-crossover phenotypes and identify the single allele that has been switched so that it is now no longer associated with its original neighboring alleles. That allele will be the one located between the other two in the sequence.

In our example y , ec , and w are on one homolog in the F_1 heterozygote, and y^+ , ec^+ , and w^+ are on the other. In the F_2 double-crossover classes, it is w and w^+ that have been switched. The w allele is now associated with y^+ and ec^+ , while the w^+ allele is now associated with the y and ec alleles. Therefore, the *white* gene is in the middle, and the *yellow* and *echinus* genes are the flanking markers.

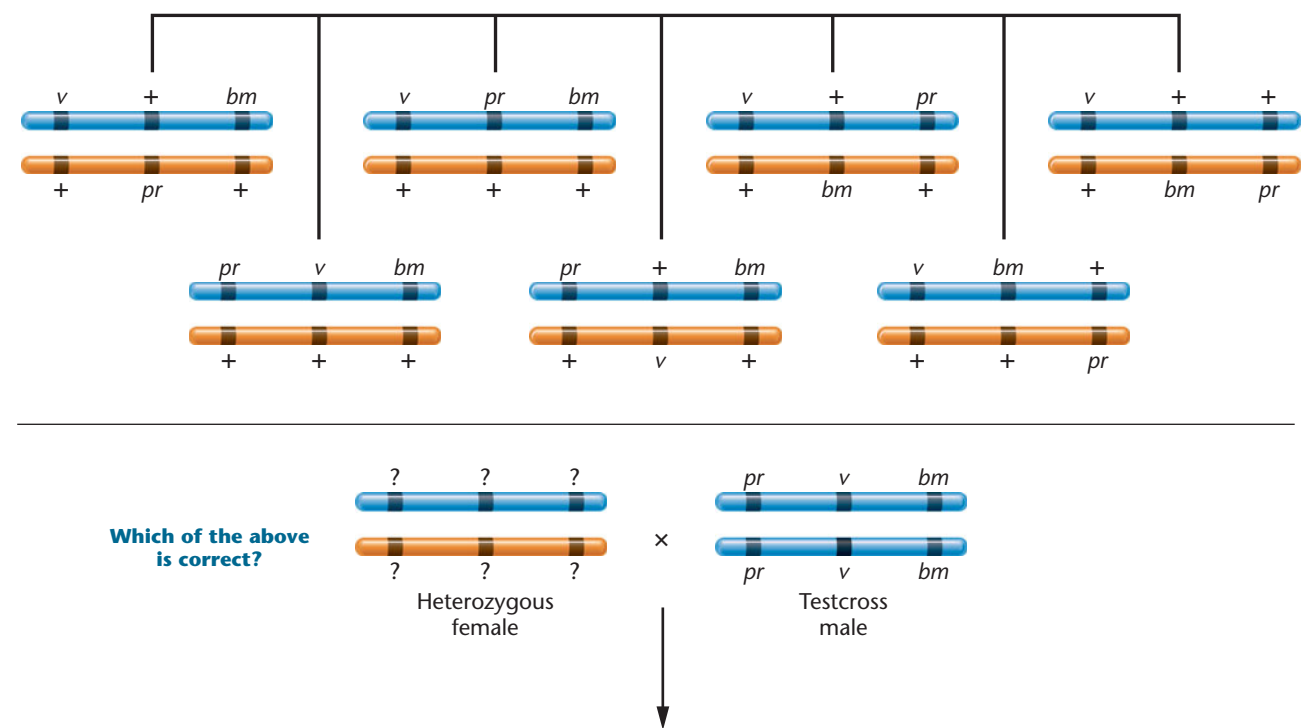
A Mapping Problem in Maize

Having established the basic principles of chromosome mapping, we will now consider a related problem in maize (corn). This analysis differs from the preceding example in two ways. First, the previous mapping cross involved X-linked genes. Here, we consider autosomal genes. Second, in the discussion of this cross, we will change our use of symbols, as first suggested in Chapter 4. Instead of using the gene symbols and superscripts (e.g., bm^+ , v^+ , and pr^+), we simply use + to denote each wild-type allele. This system is easier to manipulate but requires a better understanding of mapping procedures.

When we look at three autosomally linked genes in maize, our experimental cross must still meet the same three criteria we established for the X-linked genes in *Drosophila*: (1) One parent must be heterozygous for all traits under consideration; (2) the gametic genotypes produced by the heterozygote must be apparent from observing the phenotypes of the offspring; and (3) a sufficient sample size must be available for complete analysis.

In maize, the recessive mutant genes bm (*brown midrib*), v (*virescent seedling*), and pr (*purple aleurone*) are linked on chromosome 5. Assume that a female plant is known to be heterozygous for all three traits, but we do not know: (1) the arrangement of the mutant alleles on the maternal and paternal homologs of this heterozygote; (2) the sequence of genes; or (3) the map distances between the genes. What genotype must the male plant have to allow successful

(a) Some possible allele arrangements and gene sequences in a heterozygous female



(b) Actual results of mapping cross*

Phenotypes of offspring	Number	Total and percentage	Exchange classification
+ v bm pr + +	230 237	467 42.1%	Noncrossover (NCO)
+ + bm pr v +	82 79	161 14.5%	Single crossover (SCO)
+ v + pr + bm	200 195	395 35.6%	Single crossover (SCO)
pr v bm + + +	44 42	86 7.8%	Double crossover (DCO)

* The sequence *pr* – *v* – *bm* may or may not be correct.

FIGURE 5–10 (a) Some possible allele arrangements and gene sequences in a heterozygous female. The data from a three-point mapping cross, depicted in (b), where the female is testcrossed, provide the basis for determining which combination of arrangement and sequence is correct. [See Figure 5–11(d).]

mapping? To meet the second criterion, the male must be homozygous for all three recessive mutant alleles. Otherwise, offspring of this cross showing a given phenotype might represent more than one genotype, making accurate mapping impossible. Note that this is equivalent to performing a testcross.

Figure 5–10 diagrams this cross. As shown, we know neither the *arrangement of alleles* nor the *sequence of loci* in

the heterozygous female. Several possibilities are shown, but we have yet to determine which is correct. We don't know the sequence in the testcross male parent either, so we must designate it randomly. Note that we initially placed *v* in the middle. *This may or may not be correct.*

The offspring have been arranged in groups of two, representing each pair of reciprocal phenotypic classes. The four reciprocal classes are derived from no crossing over (NCO),

each of two possible single-crossover events (SCO), and a double-crossover event (DCO).

To solve this problem, refer to Figures 5–10 and 5–11 as you consider the following questions:

1. What is the correct heterozygous arrangement of alleles in the female parent?

Determine the two noncrossover classes, those that occur with the highest frequency. In this case, they are $+ v bm$ and $pr + +$. Therefore, the alleles on the homologs of the female parent must be distributed as shown in Figure 5–11(a). These homologs segregate into gametes, unaffected by any recombination event.

Any other arrangement of alleles will not yield the observed noncrossover classes. (Remember that $+ v bm$ is equivalent to $pr^+ v bm$ and that $pr + +$ is equivalent to $pr v^+ bm^+$.)

2. What is the correct sequence of genes?

To answer this question, we will first use the approach described in Method I. We know, based on the answer to question 1, that the correct arrangement of alleles is

$$\begin{array}{ccc} + & v & bm \\ \hline pr & + & + \end{array}$$

But is the gene sequence correct? That is, will a double-crossover event yield the observed double-crossover

Possible allele arrangements and sequences	Testcross phenotypes	Explanation
(a)	$+ \quad v \quad bm$ and $pr \quad + \quad +$	Noncrossover phenotypes provide the basis for determining the correct arrangement of alleles on homologs
(b)	$+ \quad + \quad bm$ and $pr \quad v \quad +$	Expected double-crossover phenotypes if v is in the middle
(c)	$+ \quad + \quad v$ and $pr \quad bm \quad +$	Expected double-crossover phenotypes if bm is in the middle
(d)	$v \quad pr \quad bm$ and $+ \quad + \quad +$	Expected double-crossover phenotypes if pr is in the middle (This is the actual situation.)
(e)	$v \quad pr \quad +$ and $+ \quad + \quad bm$	Given that (a) and (d) are correct, single-crossover phenotypes when exchange occurs between v and pr
(f)	$v \quad + \quad +$ and $+ \quad pr \quad bm$	Given that (a) and (d) are correct, single-crossover phenotypes when exchange occurs between pr and bm
(g)		

FIGURE 5–11 Producing a map of the three genes in the cross in Figure 5–10, where neither the arrangement of alleles nor the sequence of genes in the heterozygous female parent is known.

phenotypes following fertilization? *Observation shows that it will not* [Figure 5–11(b)]. Now try the other two orders [Figure 5–11(c) and 5–11(d)], *keeping the same allelic arrangement*:

$$\begin{array}{ccc} + & bm & v \\ \hline pr & + & + \end{array} \quad \text{or} \quad \begin{array}{ccc} v & + & bm \\ \hline + & pr & + \end{array}$$

Only the order on the right yields the observed double-crossover gametes [Figure 5–11(d)]. Therefore, the *pr* gene is in the middle.

The same conclusion is reached if we use Method II to analyze the problem. In this case, no assumption of gene sequence is necessary. The arrangement of alleles along homologs in the heterozygous parent is

$$\begin{array}{ccc} + & v & bm \\ \hline pr & + & + \end{array}$$

The double-crossover gametes are also known:

$$pr \quad v \quad bm \quad \text{and} \quad + \quad + \quad +$$

We can see that it is the *pr* allele that has shifted relative to its noncrossover arrangement, so as to be associated with *v* and *bm* following a double crossover. The latter two alleles (*v* and *bm*) were present together on one homolog, and they stayed together. Therefore, *pr* is the odd gene, so to speak, and is located in the middle. Thus, we arrive at the same arrangement and sequence as we did with Method I:

$$\begin{array}{ccc} v & + & bm \\ \hline + & pr & + \end{array}$$

3. What is the distance between each pair of genes?

Having established the correct sequence of loci as *v*–*pr*–*bm*, we can now determine the distance between *v* and *pr* and between *pr* and *bm*. Remember that the map distance between two genes is calculated on the basis of all detectable recombinational events occurring between them. This includes both the single- and double-crossover events.

Figure 5–11(e) shows that the phenotypes *v pr +* and *+ + bm* result from single crossovers between *v* and *pr*, and Figure 5–10 shows that those single crossovers account for 14.5 percent of the offspring. By adding the percentage of double crossovers (7.8 percent) to the number obtained for those single crossovers, we calculate the total distance between *v* and *pr* to be 22.3 mu.

Figure 5–11(f) shows that the phenotypes *v + +* and *+ pr bm* result from single crossovers between the *pr* and *bm* loci, totaling 35.6 percent, according to Figure 5–10. Adding the double-crossover classes (7.8 percent), we compute the distance between *pr* and *bm* as 43.4 mu. The final map for all three genes in this example is shown in Figure 5–11(g).

NOW SOLVE THIS

5–3 In *Drosophila*, a heterozygous female for the X-linked recessive traits *a*, *b*, and *c* was crossed to a male that phenotypically expressed *a*, *b*, and *c*. The offspring occurred in the following phenotypic ratios.

+	<i>b</i>	<i>c</i>	460
<i>a</i>	+	+	450
<i>a</i>	<i>b</i>	<i>c</i>	32
+	+	+	38
<i>a</i>	+	<i>c</i>	11
+	<i>b</i>	+	9

No other phenotypes were observed.

- What is the genotypic arrangement of the alleles of these genes on the X chromosome of the female?
- Determine the correct sequence and construct a map of these genes on the X chromosome.
- What progeny phenotypes are missing? Why?

HINT: This problem involves a three-point mapping experiment where only six phenotypic categories are observed, even though eight categories are typical of such a cross. The key to its solution is to be aware that if the distances between the loci are relatively small, the sample size may be too small for the predicted number of double crossovers to be recovered, even though reciprocal pairs of single crossovers are seen. You should write the missing gametes down as double crossovers and record zeros for their frequency of appearance.

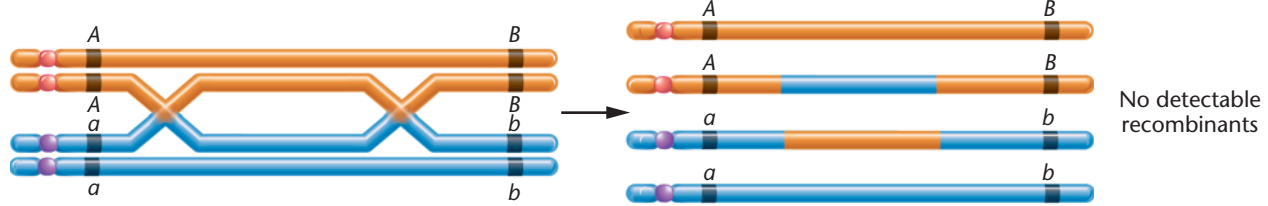
5.4

As the Distance between Two Genes Increases, Mapping Estimates Become More Inaccurate

So far, we have assumed that crossover frequencies are directly proportional to the distance between any two loci along the chromosome. However, it is not always possible to detect all crossover events. A case in point is a double exchange that occurs between the two loci in question. As shown in Figure 5–12(a), if a double exchange occurs, the original arrangement of alleles on each nonsister homolog is recovered. Therefore, even though crossing over has occurred, it is impossible to detect. This phenomenon is true for all even-numbered exchanges between two loci.

Furthermore, as a result of complications posed by multiple-strand exchanges, mapping determinations usually underestimate the actual distance between two genes. The farther apart two genes are, the greater the probability that undetected crossovers will occur. While the discrepancy

(a) Two-strand double exchange



(b)

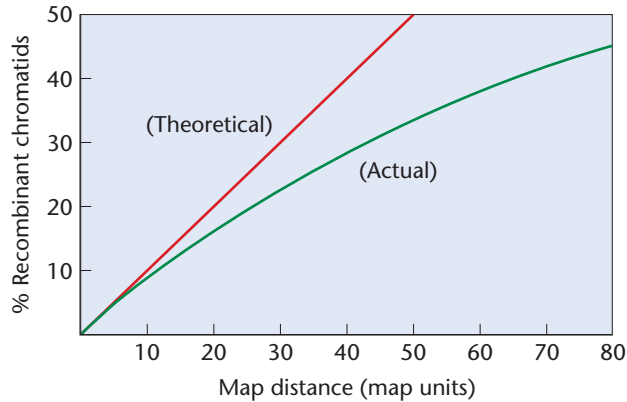


FIGURE 5-12 (a) A double crossover is undetected because no rearrangement of alleles occurs. (b) The theoretical and actual percentage of recombinant chromatids versus map distance. The straight line shows the theoretical relationship if a direct correlation between recombination and map distance exists. The curved line is the actual relationship derived from studies of *Drosophila*, *Neurospora*, and *Zea mays*.

is minimal for two genes relatively close together, the degree of inaccuracy increases as the distance increases, as shown in the graph of recombination frequency versus map distance in Figure 5-12(b). There, the theoretical frequency where a direct correlation between recombination and map distance exists is contrasted with the actual frequency observed as the distance between two genes increases. The most accurate maps are constructed from experiments in which genes are relatively close together.

Interference and the Coefficient of Coincidence

As review of the product law in Section 5.3 would indicate, the expected frequency of multiple exchanges, such as double crossovers, can be predicted once the distance between genes is established. For example, in the maize cross of the previous section, the distance between *v* and *pr* is 22.3 mu, and the distance between *pr* and *bm* is 43.4 mu. If the two single crossovers that make up a double crossover occur independently of one another, we can calculate the expected frequency of double crossovers (DCO_{exp}) as follows:

$$\text{DCO}_{\text{exp}} = (0.223) \times (0.434) = 0.097 = 9.7\%$$

Often in mapping experiments, the observed DCO frequency is less than the expected number of DCOs. In the maize cross, for example, only 7.8 percent of the DCOs are observed when 9.7 percent are expected. **Interference** (*I*), the inhibition of further crossover events by a crossover

event in a nearby region of the chromosome, causes this reduction.

To quantify the disparities that result from interference, we calculate the **coefficient of coincidence** (*C*):

$$C = \frac{\text{Observed DCO}}{\text{Expected DCO}}$$

In the maize cross, we have

$$C = \frac{0.078}{0.097} = 0.804$$

Once we have found *C*, we can quantify interference (*I*) by using this simple equation

$$I = 1 - C$$

In the maize cross, we have

$$I = 1.000 - 0.804 = 0.196$$

If interference is complete and no double crossovers occur, then *I* = 1.0. If fewer DCOs than expected occur, *I* is a positive number and **positive interference** has occurred. If more DCOs than expected occur, *I* is a negative number and **negative interference** has occurred. In this example, *I* is a positive number (0.196), indicating that 19.6 percent fewer double crossovers occurred than expected.

Positive interference is most often observed in eukaryotic systems. In *C. elegans*, for example, only one crossover event per chromosome is observed, and interference along each chromosome is complete (*C* = 1.0). In other organisms, the closer genes are to one another along the chromosome,

the more positive interference occurs. Interference in *Drosophila* is often complete within a distance of 10 map units. This observation suggests that physical constraints preventing the formation of closely spaced chiasmata contribute to interference. The interpretation is consistent with the finding that interference decreases as the genes in question are located farther apart. In the maize cross illustrated in Figures 5–10 and 5–11, the three genes are relatively far apart, and 80 percent of the expected double crossovers are observed.

5.5

Drosophila Genes Have Been Extensively Mapped

In organisms such as fruit flies, maize, and the mouse, where large numbers of mutants have been discovered and where mapping crosses are possible, extensive maps of each chromosome have been constructed. Figure 5–13 presents

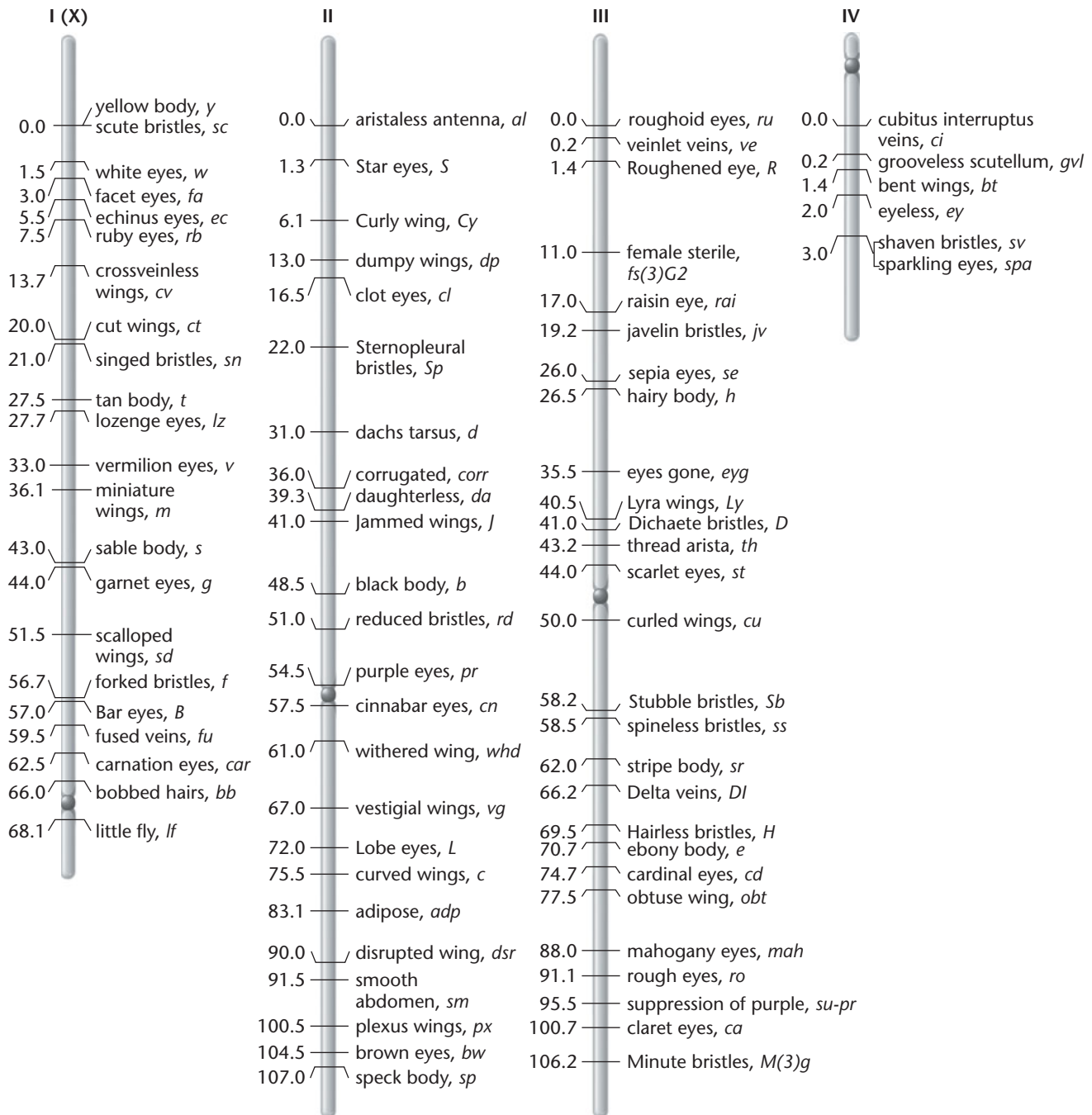


FIGURE 5–13 A partial genetic map of the four chromosomes of *Drosophila melanogaster*. The circle on each chromosome represents the position of the centromere. Chromosome I is the X chromosome. Chromosome IV is not drawn to scale; that is, it is relatively smaller than indicated.